Department of Information Technology

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch	**:** IV Semester B.Tech IT
Course Code / Title	**:** CS3491/Artificial Intelligence & Machine Learning
Name of the Faculty member	**:** Mr. G. Mareeswari

Innovative Practice Description

- **Unit / Topic: Unit IV / Ensemble Techniques and Unsupervised Learning /** Expectation Maximization
- **Course Outcome:** CO4
- **Activity Chosen:** Flipped Classroom
- **Justification:**

  The flipped classroom is an instructional strategy where traditional teaching methods are inverted or "flipped." Students are assigned materials to study before the class. This activity allows them to gain exposure to the content independently at their own pace. Classroom time is then dedicated to interactive learning activities, discussions. It assists the instructor/faculty in redefining in-class activities, assigning homework, and maintaining student interest in the Expectation Maximization concept.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**
  - The learning materials were posted to the students in canvas two days before the topic discussion.
  - The students already learned the concept of other ensemble techniques and unsupervised learning in the previous class
  - The students are instructed to study the material and take notes on the topic based on their understanding.
  - On the day of the activity the students were grouped into a team of 2 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes.

- After the discussion one member from each team had a chance to present and share the idea of Expectation Maximization to the entire class as shown in Fig.1 & 2
- **Plan:** Separate student's teams are formed and made them to explore as a group.
- **Identify and Share:** Related materials are identified and posted relevant materials/notes of Expectation Maximization topic. Aarthi. S, Harikrishnan. K and Dinesh. S are requested to present the concepts they have learnt and share it to the entire class students.
- **Evaluate:**
  - I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class.
  - Make the students to find out the answer by their own by learning.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO4	**2**	**2**	**2**	**1**	**1**	**2**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PSO1**
	2	**2**	**2**	**1**	**1**	**2**
Justification for correlation	Apply basic Knowledge and fundamentals in ML to understand the EM concept	Using the concept EM we can able to analyze and identify complex Engineering problems.	Able to design and develop solutions for various complex problem.	Students involved in this activity as an individual and also as a team.	Communicate/ share the ideas with other students	Ability to find solution for data science problem.

- **Images / Screenshot of the practice:**

Figure 1 and 2 shows the Glimpses of presentation done by Aarthi. S , Harikrishnan. K and Dinesh. S during the flipped class room Activity.

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Student felt good, since they can study at their own pace/time.
    - They felt that through this type of active learning, they can explore themselves more.
  - ***Benefit of the practice:***
    - This activity provides one-to-one time between teacher and student.
    - It represents the way of learning by collaboration within the students.
    - Students learn at their own pace.
    - Practical things – like missing class due to illness – become less problematic.
    - It encourages students to come to class prepared.
  - ***Challenges faced in implementation:***
    - The activity depends on student preparation – few students did not come prepared.
    - The depth of the subject can be dictated by the student themselves or the group the student is working with.
    - The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.
- https://en.wikipedia.org/wiki/Flipped_classroom
- youtube.com/watch?v=BCIxikOq73Q
- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model

Signature of Faculty Member **HOD**